

Follow up_ Aim 1 Inhibitory neuron experiments

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SUMMARY KEYWORDS

Inhibitory neurons, GABA agonist, potassium titration, cyclic AMP, force blend, proprietary media, neuron plasticity, EPSPs, IPSPs, PAMs, neuron density, potassium methyl sulfate, imaging buffer, action items.

SPEAKERS

James Fink, Speaker 1, Speaker 2, Speaker 4, Speaker 3, Steven Ryan



James Fink 00:00

Meeting earlier today, Steve to talk about, like experiments in culture, and otter was there, and then we left the meeting. And then two and a half hours later, three hours later, kathiann sitting over in the lobby and talking. She had a computer. I was actually on otter, and I'm looking at the otter homepage, and like the meeting from three hours ago is still running. It's seeing the words I'm speaking in the lobby like copy a picture, pretty sure your meeting is still on. Three hours ago, you went some shit about me.



Steven Ryan 00:33

It's just been following her around all day.



James Fink 00:38

This is easier for me. Do I just plug it, plug it in, and then go up to next to your microphone and hit that down arrow, and then should pop up with Jabra. For some reason, I just think of jar, jar banks when I think of Jabra, for some reason, what's the job jaw? And it has an R jar job, yeah, I'm saying it's weird. It just that's what needs to think of as B branch versus job, bra. It's how my mind works. We can I got it little ears and a tongue. That's right. I have watched all Star Wars, but supposedly drugs are basically a bad guy, or like, people hate him something people hate him. Yeah, because he does something bad. He's like, he's he becomes bad.



Steven Ryan 01:27

He reminded people that Star Wars movies are for children, which which people hate because they wanted them to be for them.



James Fink 01:35

I also thought that baby yoda's name was baby Yoda. Until recently, someone called him, like, jewelry or something like that.



Steven Ryan 01:44

And that's like, yeah, like the whole person. For the whole first season of that TV show, they didn't tell you his name, so people called him baby Yoda. And then in the second season, you find out he has a name. But apparently, like, the director of the show, and the guy who does all the writing gets really pissed if people call him baby Yoda. And it's like, well, like, yeah, what Did you expect was gonna happen? I



Steven Ryan 02:20

time Star Wars better show up in the action items for this meeting.



James Fink 02:23

Well, you sent to Steve Kathy, sent me action items from our last year. Last meeting is all about our skeptics, our kids and friends. Skepticism with Santa. I was



02:33

looking for any between neurons, and this is what

J James Fink 02:40

so we have so there's couple things to outline here. Kathy is planning an experiment with Luis to test some interventions to enhance sort of like gap of maturation and culture on culturally, purely culture side, I think we it's a gray area. I think you and I, Steve, can help design experiments in that domain as well. That's not like, really Luis's. I mean, his is mostly the stem cell stage. But some experiments can't be is working on do involve, like, media that we purchase and stuff. So there's an experiment where thermo has a proprietary, a new, proprietary media they're not selling. They we've sort of signed up to test it out and share the data with them to show in fact, it's like better. We think it could be better by enhancing inhibition, because our cells right now, you know, are lacking in that, so that'd be really great data for them. So it's a win, win, win that works. We start planning that experiment. But another one is in some of the journal clubs Kathy presented, they use force blend to help boost cyclic AMP and sort of get these cells along.

S Steven Ryan 03:47

This is like culture media, yes.

J

James Fink 03:52

Well, it's just forcefully, forcefully in plus roll up RAM. So roll up programs a significant phosphodiesterase inhibitor. So actually maintain force, Glen will up regulate sickly and pee. But cells are very smart, and they'll immediately sort of degrade it. So probably few weeks of force good is not going to kill the cells. But I can promise you, when I do plasticity experiments, and we will do these in different aim, Steve, when you expose cells, neurons, human neurons, to force Glen, plus roll up RAM and zero magnesium. It comes so excited, toxic, toxic that if you do that for like, more than a half hour, they all die. So 15 minutes is enough to, like, induce plasticity. They'll be really healthy. They'll have crazy amounts of activity. And then you wash it out, and activity will maintain for, like, over an hour. It'll start to come down, but it maintains. But if you I've gone away and, like, done some stuff and got lunch at work and came back and like, oh shit, I forgot to turn the Forgot to watch this thing out that all the cells are dead. So probably not, probably states, if it's just forcefully. So that's one experiment. The other thing that we had talked about, speaking of good ideas from Gemini, when I was asking questions, Steve, I think you're part of some of this. Gemini suggesting one of the hypotheses. I fitted Hong Kong's deck. And said, What could explain some of this stuff? And actually one of the theories. And then I took a screenshot of some seek biz data, and I said, Oh, this is really interesting. Gabriele, a one Alpha One very low, and alpha three is very high, which is like, more mature, and it counts for slower things. So actually, when we were doing some of this with Luis, it suggested that we actually apply chronic GABA agonist in culture and incubator into tricking the cells that GABA is there all the time, right? Hey, you're mature, because gap is around, right? So we're gonna actually include, use them all, or some similar compound for, like, full 30 days, or like, whatever, two weeks, something. So that's one experiment that's happening. The stuff I'd like to do today is take the list that I've actually as Kathy sent and to propose one with a couple variables, or two experiments with a few variables each that would act as like the follow ons to the proprietary media from thermo and the miso mal force Glen experiments. The things we had discussed last time were the potassium titration with the goal of not just showing upsp, Steve, but like to show a titration of reversal potential. Like, I think it'd be really nice data. Like, instead of doing current clamp and seeing the reversal, you just do dose responsive. That's exactly what I imagine when you describe that experiment. PSPs, media. Upsps, little, tiny, upsp, flat, and then reversal. That would be one. The second, I think we just discussed, is gabapentin, so that benzodiazepines and some others, where I see that going is like Hong Kang detected some bare bones, number of ipsp, and some number that, I hope you find out tomorrow, EPSPs, that was one of his action items, a PAM would both increase the number because they would get bigger. So if they're thinking the noise, they should get larger, so easier to detect. So I guess, my guess is, like you get more ipsp and more UPS piece, they probably aren't sufficient to solve our problem for drug discovery, but it would be moving in the right direction. Like, okay, Pams. Like, do this stuff is actually GABA mediated, because they're getting bigger, both upsp and upsp when we use Pams. So those are two of the obvious ones. You have smart things to play around with. It'd be like, GABA neuron density. Ideally, we do like physiological densities, because that's what it is in the brain. But you can imagine scenarios where, like, if you have less inhibitory cells, then maybe they more strongly connect to each excitatory neuron, just by nature of there being fewer of them, less competition. Or that's the reverse. You have to have way more inhibitory neurons, and then you get better inhibition, I don't know. So you can imagine playing with the ratio. Those are some of the thinking as a prereq to this meeting. But see what's your like highest rated parameters that you would give a go. Ah,



James Fink 08:23

we discussed Gavin sniffers. Kathy had mentioned it to me in passing a little earlier today. Is that just a matter of ordering of construct and putting them in ourselves? Steve, I don't



Steven Ryan 08:34

actually know that much about them. I don't know. Let's see what's Jim and I say,



Steven Ryan 08:48

I don't like if there are just traditional fluorescent reporters. I don't know if they need to be expressed. I think they might actually just be like aqueous,



08:56

right? So you



James Fink 08:58

just put them in,



Steven Ryan 08:59

yeah, but I don't know if we'll have the illuminators and filters for them. That would be the



James Fink 09:05

there, so maybe put those in the back burner. So can me, can you pull up a plate layout and we can least play with a few of these parameters we talked about the start. I



James Fink 09:28

i am i got to spot one of the robot dogs about an hour ago, threw me for a loop lobby, sitting there drinking my coffee.



09:46

Okay? You



09:47

want me to put it? Yeah? Sure you



James Fink 09:56

can start filling it out. So there's 61 layout. Okay? I



James Fink 10:32

All right, so let's start the potassium chloride. So we don't need to compare these different conditions to each other. They're all independent, and they may be additive. So the follow up experiment might say, Let's combine two feet. But I think the sort of we should treat each of these sort of separate, at least most of them we can we can change as we go. So the first one would be like a, let's say, potassium chloride concentration response. Now, Steve, and this is a question I can ask Gemini after this meeting, which is like, the problem with actually changing KCL is that you're also changing chloride, which is the egg I would conduct. So there might be an alternative to KCL to change the membrane potential, but not change the chloride reversal.



Steven Ryan 11:22

Yeah. I mean, what this? This is something that we will be doing in the imaging buffer, right, not in the culture, yeah. Can we tweak the composition of the imaging buffer to compensate for the change in chloride concentration, right?



James Fink 11:38

Once you run an experiment where we you change the membrane potential of



Speaker 1 11:50

a neuron in dose response by

J James Fink 11:57
altering external k plus concentration, so we can

S Speaker 1 12:07
study the reversal of GABA mediated ipsp's and EPSPs in our cultures,

i 12:21
the norm normally

S Speaker 1 12:23
Use KCl, but of course, for is,

S Speaker 1 12:37
yeah, let's see. Well, the answer

J James Fink 12:43
to manipulate the membrane potential via external potassium while keeping the chloride concentration constant, you need to use a non permeable or inert potassium salt. Using KCl will confound your result, because gap or something primary, permeable chlorine, change in chlorine would shift the chlorine equilibria potential, making it possible to help the shifts is due to change in voltage. Here's our choices. Potassium methyl sulfate, commonly used with high precision electrophysiology, high ionic mobility, minimizes the liquid junction potential and very low permeability through most channels. Potassium gluconate, standard patch clamp, very common and stable. However, gluconate can weakly buffer calcium and has lower mobility than methyl sulfate. Potassium acetate reliable for changing the end but acetate is slightly more metabolic permeable than gluconate or methyl sulfate, and finally, potassium methane sulfonate, similar to methyl sulfate, often used interchangeably, although methyl sulfate is sometimes preferred for healthy looking cells in long recordings, yeah, we

Steven Ryan 13:45

used to use potassium gluconate in our patch solution, but I don't think we typically used it in The ACSF.

J

James Fink 13:53

Are all those salts

13:58

on in ACS

14:03

no



14:06

internal solutions. Let's see.

J

James Fink 14:11

Says here, when, when, when increasing K^+ external using K_m , you can must decrease the concentration of sodium chloride in your bath solution by an equivalent solution by an equivalent amount to keep the total osmolarity constant, calculate the E_{Na} with the Nernst equation. I'll let you do that, Steve, to predict your target membrane potential using the Nernst equation. Potassium at room temperature, this simplifies to the Nernst potential when switching between different external salts. Beware the K_m so four and K^+ glutamate have different junction potentials compared to K^+ chloride. You may need to correct your multiple endings matter for us because E_{Na} is typically close to E_{K} . Keeping E_{K} outside stable is critical in your cultures that you chloride cells by increasing K^+ , you can observe transition of E_{Na} responses from hyperpolarism. Deep order into this transition point is escapable. Okay. While these salts are staples of internal pipette solutions, using them an external ACSF solution require some care. Your right to be cautious. Shifting the dominant anion in the bath is more aggressive. Change the cells environment. They compete inside of pipette.



15:21

Here's the breakdown,



James Fink 15:24

the big 3k meso, 4k gluconate. And chemist me so 3k meso, four is marked as excellent for external stability, high mobility. Does not bind calcium. Best choice for your experiment, the biggest risk when using potassium gluconate is in the bath. Is its affinity for divalent cations such as K calcium. You replace 50% of your external NaCl K glucanate, the effective concentration of free calcium your ACF will drop significantly.



15:56

So that's a no so



James Fink 16:03

so maybe Kathy mentioned potassium and the so 4k and the so four as an ion to try in dose response. So in action, I'm here Kathy is to get the media recipe from Jen, and we're going to feed that to because I don't like neuron stomach, I'm allergic, and it gives me PTSD. We're going to feed that to Gemini, and we're going to ask it to propose a dose response experiment, and what, what millimolar of potassium, methyl sulfate to use, and what to reduce the NAC and a sodium chloride plan that makes sense.



Speaker 2 16:43

Okay, we need a recipe for that potassium sulfate, whatever. Ask Gemini for experiments.



James Fink 16:50

But for me, just get it from Jen that the recipe. Okay, she'll have it like probably find a bit easier. She just sends it to me. Okay, that's, that's number one. And Steve, I think six point CRC for this is probably sufficient. Like, if the dynamic range probably not huge here,

S Steven Ryan 17:10
yeah, I would think, yeah.

J James Fink 17:13
Or three, three concentrations.

S Steven Ryan 17:18
I mean, do we want to go, I guess, like, do we want to go up and down from our normal

J James Fink 17:21
concentration? Yes. So the normal, be it center, yeah.

S Steven Ryan 17:26
So, I mean, you could get by with three. Then you just want to make sure that the two, the two perturbed concentrations, are like, pretty different. What's our what's our normal?

J James Fink 17:37
KCL, yeah. So I can just find this on now, buffer. Let's type that in to see what we get. Oh, here he is. First thing that pops up. That's great.

J James Fink 18:02
Brain Fizz, stem cell 1x what is a brain physics? Q state. This is not the recipe. This is the tracker

S Speaker 1 18:16
buffer recipe. I we have,



18:26

we all have a recipe. Yes, we do. We do, buffer prep, SOP, Otter imaging department, SOP, work instructions,



James Fink 18:38

lab departments,



18:41

SOPs and Imaging, imaging plans



Speaker 1 18:47

and results and the buffer prep, SOPs and working stuff, obviously, okay. And then which one SOPs and templates, proper? Prep? Yes, okay. Is it is



James Fink 19:05

1.1183 grams. Doesn't tell us the concentration, though.



James Fink 19:17

Yeah, I don't think we know. Steve,



Steven Ryan 19:19

all right, so we should work out with the millimolar concentration of potassium is, but like, Yeah, something like half and double that is probably enough to get a change.



James Fink 19:33

So we should download this. We're going to want to, oh, but that's the problem. Is that's brain foods. We need tyroids.

 Steven Ryan 19:42
Oh, is that the imaging buffer for these?

 James Fink 19:47
Yeah, for synaptic I don't see a tie Rhodes.

 Speaker 2 19:52
That's yeah protocol for making plain brain force or tyroids imaging. There's all of the

 Speaker 3 19:59
same one, not where I am in work. Instructions and SOPs. It says buffer, prep, SOP, draft, v6
Yes, 2022, it says tie roads in here, yeah.

 20:11
It says brain first or,

 20:18
well, they can't be the same, all

 Steven Ryan 20:21
right. Well, tie roads, at least, is fairly standard.

 James Fink 20:24
It's 5.4 to make 10 liters. Tid roads, we have each of the following using a scale. The new blues,
alright. So there it's

 Steven Ryan 20:34

all right. So good. The good news, the good news with tie rodes is, if we just change the potassium chloride, so there you're getting chloride, mostly from the sodium chloride. So if you change the potassium chloride, it's not really going to change the CL concentration at all. So we can just go down to like two. And maybe we should check literature. We can, like, go down to two and up to like eight or 10.

 James Fink 20:58

Okay, so we will what's planned for that. So maybe Kathy, if you go to the plate layout, just do three, three buckets of wells. We can do the same as we did for spontaneity of the bursting, which is like one row, standard, low, high, standard, low, high, standard, low, high.

 21:19

Okay, that's good.

 James Fink 21:26

What are the best?

 James Fink 21:33

So one's going to be diazepam. So we should check with Ben if we have that

 James Fink 21:49

ambient Steven, propofol. Have you ever used that?

 Steven Ryan 21:59

Not in an experiment? Pretty sure I got that in the dentist

 James Fink 22:03
once said the barber call.

 Steven Ryan 22:08
I think we used to use that to kill rats, but no, I don't think I've ever used it for in, like a, like a Ephesus experiment.

 James Fink 22:15
Okay, so we should ask Ben for those which wouldn't Dart, yeah, that we definitely have, or we've had in the past. I've used it here for other stuff. I'm gonna send this to the chat. The other two are barbiturates, which should also be something we can order.

 22:37
Zilp to them. This

 James Fink 22:41
is interesting. So that's highly, highly specific for alpha one, which we think doesn't exist. So we should try that as well. So me, this is experiments. Kathy is a lower it's like this. We just do this one separately because we're probably gonna want to order these drugs prep them. So maybe this is the second experiment, instead of at the same time,

 23:08
krm 281

 James Fink 23:11
a research grade Canada avoids a sedation associated alpha one use for circuit level studies. Have you ever heard of that? Steve, what's it called?



23:20

Krm 281 Nope. That one's new. Must have all these, and we can't spend over all of them. MP, 302,



23:35

nope, that one's new to five. Pam,



James Fink 23:40

potent, selective, positive, allosteric, modular of gaviar cipher is contained alpha five subun with a K_i value, 55 nanomolar, 15, six and 12 fold selectivity of Alpha one Alpha All right,



James Fink 24:00

so we should work. We should work. We should work all these really do, like, serious gap experiments. I think we should explore some of the biology here. Yeah, like, if there's a perception unit composition, like matches the functional or say, hey, three is high, one's low, and then the one Pam doesn't act as good as the two, three Pam, that's telling that checks out. Yeah. I think that would be like a good story to tell.



James Fink 24:35

All right. Other Other thoughts, maybe that everyone can do Kathy as a ratio experiment. Okay?



Speaker 4 24:55

Gabba, hmm.

 James Fink 25:04

So here's a thinking, and we can just, why don't we? I'll meet a scene from four to 430 so you guys can learn stuff done. So the last action, I would say, is once you put a plate layout together for what ratio experiments look like. You can also do maybe a three ratios. I feel like anything more granular than that's kind of useless. Like, what's the difference between like 6040, and 7030, probably not a huge difference.

 25:37

We did that. I think I know, but

 James Fink 25:41

I'm thinking of doing that with the PAM, or even with the potassium, because then, like, yes, by itself, is probably not going to do anything but, but if you are able, if we're able to evoke some more gather than we normally see, then we can, then we can actually say, okay, Hong Kang detects 13 ipsp's in the baseline conditions and 40 EPSPs, which is barely anything but, but now we've changed the ratio. It's either largely stayed the same or we doubled it or halved it, you know, I mean, like I can see that being

 26:12

the case,

 James Fink 26:16

part of that study, when you do the design of that, you should include a no GABA, no inhibitory neurons, Ng and two only, and we'd still run GABA assay on Ng and two with blocking all the receptors, just to show it's not like a sheriff artifact or something like that, I think that would be a good control.

 26:38

Okay. Is that good for now?

J James Fink 26:40

So I proposed for next steps is maybe just get the small group together Friday and we can actually finalize, like, what the design looks like. In the meantime, check a bend on ordering those drugs.

S Speaker 3 26:53

Yeah, already. Thank you so much. Thank you, Jamie. Let's jump to the other the other call, yeah,

S Steven Ryan 27:02

yeah, thanks, everyone. Bye.